

ARKOPROBHO MITRA

AI Engineer · LLM Systems · Agentic Pipelines · AWS Certified Cloud Practitioner

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PROFESSIONAL SUMMARY

AI Engineer with **5+ years** at Infosys, with a track record of shipping production **GenAI** and **ML systems** at Infosys Topaz. Currently leading sub-teams across banking **RAG**, insurance claims intelligence, and multi-agent procurement automation. Background in operational data systems such as **PL/SQL**, **Elasticsearch**, **QlikSense**, and **ServiceNow**, across large-scale logistics workflows. B.E. from Jadavpur University, **AWS Certified Cloud Practitioner**, and co-author of a Springer-published research paper on multilingual OCR. Builds AI systems engineered for production — cost-optimised, secure, and observable from day one.

TECHNICAL SKILLS

- **AI / LLM Frameworks:** LangGraph, LangChain, LangFuse, FAISS, RAGAS, CrewAI, MCP, Pydantic AI, Anthropic SDK, OpenAI SDK, LiteLLM, Prompt Engineering, Safety Filters.
- **Machine Learning:** pandas, XGBoost, LightGBM, scikit-learn, SHAP, LayoutLMv2, RoBERTa, SpaCy NER, Evidently AI, MLflow, Azure ML, CI/CD Pipelines, Observability.
- **Backend:** Python, FastAPI, asyncio, asyncpg, Spring Boot 3 (Java 17), Node.js, APScheduler, SQLite, PostgreSQL
- **Frontend:** React 19, Redux Toolkit, Vite, Tailwind CSS, JavaScript, Google Apps Script
- **Database & Search:** PL/SQL, Elasticsearch, Azure AI Search (Cognitive Search), PostgreSQL, SQLite
- **Cloud & Infrastructure:** AWS EC2, Azure OpenAI, Azure Document Intelligence, Azure Key Vault, Azure Blob Storage, Docker, Vercel, GitHub Actions, Cost Optimizations.
- **Enterprise Tooling:** QlikSense, ServiceNow, SAP Ariba REST API, SharePoint REST API

WORK EXPERIENCE

AI Engineer / Sub-team Lead · Infosys Topaz (Applied AI Practice) · Kolkata, India

Nov 2023 – Present

► Project 3 · Multi-Agent Procurement Copilot (Aug 2025 – Present)

- Leading a team of 3 AI Engineers to architect and ship a **LangGraph**-orchestrated multi-agent system for a global manufacturing enterprise, delivering a **35% reduction in procurement cycle time** and cutting RFQ turnaround from **3 days to 4 hours**.
- Architected a 5-agent graph (Planner, Supplier, Contract, Compliance, Synthesis) integrating **SAP Ariba**, **ServiceNow**, and **SharePoint REST APIs** via **LangChain** tool definitions.
- Implemented vector-backed contract **RAG** on **Azure AI Search** with metadata access-control filtering; blue-green index deployments with canary health checks eliminate silent data-loss incidents.
- Architected enterprise-grade security posture: zero-secret agent design via **Azure Key Vault** and **Managed Identity**; all write operations gated behind HITL approval nodes; immutable audit logging to **Azure Blob Storage** for GDPR/DPA compliance.
- Evaluated system across three levels: tool selection accuracy (**> 90%**), workflow completion rate (**> 95%**), and **RAGAS** faithfulness + SME scoring per sprint.

► Project 2 · Insurance Claims Intelligence Platform (Nov 2024 – Sep 2025)

- Spearheaded a team of 4 AI Engineers to design and deploy an end-to-end claims intelligence platform for a global insurer, achieving a **40% reduction in processing time** across **OCR**, field extraction, fraud detection, and HITL routing.

- Deployed **Azure Document Intelligence** + **LayoutLMv2** classifier (**94% accuracy** on **15K labelled pages**) for document-type routing across 12 classes (FNOL forms, medical records, police reports, etc.).
 - Built **XGBoost** fraud-detection model (**800K claims**, **3% fraud prevalence**) with **SHAP** explainability for audit-readiness; handled class imbalance via `scale_pos_weight` rather than SMOTE.
 - Integrated **Evidently AI** drift monitoring triggering automated **MLflow** retraining; moving from **5-day manual SLA** to automated overnight retraining when PSI exceeds threshold.
- **Project 1 · Enterprise Knowledge Assistant — Global Banking (Nov 2023 – Dec 2024)**
- Sub-team Lead (2 engineers): delivered a 2-phase internal **GenAI** assistant over a **2.5M-document corpus** for a North American investment bank, reducing compliance query time by **60%** at **30,000+ monthly active users**.
 - Phase 1: Custom **Python** orchestration over **Azure Cognitive Search** (BM25 + Semantic Ranker), **Azure Form Recognizer OCR**, and **GPT-3.5-Turbo** on **Azure** for answer synthesis.
 - Phase 2 (Topaz): Migrated to hybrid vector + BM25 retrieval (Reciprocal Rank Fusion); **+14% precision** on regulation-code queries. Upgraded to **GPT-4**; introduced **LangChain** post enterprise security review.
 - Enforced role-based document access via pre-retrieval metadata filters on Azure AD group memberships; model never received restricted content.
 - Built **200-query golden eval dataset** with compliance SME; embedded evaluation gates in CI/CD; adopted **RAGAS** in Phase 2 for automated scoring.

Systems Engineer — Data & Logistics · *Infosys (Logistics & Data Systems Practice)* · Remote / India Aug 2021 – Oct 2023

- Maintained and optimised **PL/SQL** stored procedures, queries, and data pipelines supporting daily logistics operations across large-scale shipping and supply chain workflows.
- Administered **Elasticsearch** indices for real-time shipment tracking and operational search; tuned mappings and queries for throughput and latency.
- Built and maintained **QlikSense** dashboards surfacing operational KPIs for logistics and operations teams.
- Managed **ServiceNow** workflows for incident and change management, integrating with client legacy systems and internal databases.
- Collaborated with cross-functional teams on client system integrations and data migration tasks, supporting scalable data architecture across the logistics practice.

PERSONAL PROJECTS

Agentic Job Application Pipeline 2026
LangGraph multi-agent system that scrapes job listings, scores them with an **LLM**, drafts tailored cover letters, and tracks application state — full agentic automation of the job search workflow.

LangGraph Fact-Checker with Human-in-the-Loop 2026
 Multi-node **LangGraph** graph with web search tool, claim verification agent, and HITL approval step; **Langfuse** observability integration for trace-level evaluation.

Full-stack E-Commerce Platform with RAG Assistant 2025
 Microservices architecture: **React 19** + **Redux**, **Spring Boot 3** (Java 17), **FastAPI** + **asyncpg**, **Node.js** semantic search (@xenova/transformers). **RoBERTa** + **XGBoost** review sentiment pipeline; **Gemini 1.5 Flash RAG** assistant; distributed JWT auth across all 3 services.

EDUCATION

Bachelor of Engineering (B.E.) 2017 – 2021
 Jadavpur University, Kolkata, India

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services

How to handle bi/tri-lingual Indic texts in a single image? A new dataset of natural scene and born-digital images

Multimedia Tools and Applications, Springer · Vol. 81, pp. 15367–15394 (2022)

Department of CSE, Jadavpur University · [DOI: 10.1007/s11042-022-12596-7](https://doi.org/10.1007/s11042-022-12596-7)

Research scope: constructed and published a novel multilingual dataset of natural scene and born-digital images containing Indic script combinations, advancing **OCR** and scene-text detection research for low-resource scripts.